

A Study on Diffusion Models

From Stochastic Differential Equations to the Probability-Flow ODE



Jangha Lee

Seoul National University | 서울대학교

August 14, 2026

Outline

What this talk covers

01

Overview

The road from an SDE to score-based sampling

02

SDE

ODEs driven by Brownian motion & white noise

03

Examples

Brownian motion with drift; the OU process

04

Transition

Marginal & conditional laws; why “diffusion”

05

Generator & FP

How the probability density evolves in time

06

PF-ODE

A deterministic sampler & the score function

SECTION 01

Overview

What we learn, and why we learn it

01

The Big Picture

why an SDE?

SDE

Model data as a diffusion process driven by noise.

$$dX_t = f(X_t, t) dt + g(t) dW_t$$



Fokker–Planck

Track how the density evolves in time.

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g^2 \Delta p_t$$



PF-ODE

Deterministic sampler via the score function.

$$\dot{X}_t = f - \frac{1}{2} g^2 \nabla \log p_t$$

Why it matters

Diffusion models generate data by reversing a noising SDE. Understanding the SDE → its density → an equivalent deterministic ODE is exactly the theory behind **score-based generative models**.

- What is the stochastic process $\{X_t\}$? — probability & Brownian motion
- How does its density $p_t(x)$ evolve? — the Fokker–Planck equation
- Can we sample deterministically? — the probability-flow ODE

SECTION 02

Stochastic Differential Equations

ODEs driven by Brownian motion

02

What is an SDE?

definition

Stochastic Differential Equation

An SDE is an ODE with randomness injected through its coefficients.

$$dX_t = b(X_t, t) dt + \sigma(X_t, t) dW_t$$

Equivalently, in integral (Itô) form:

$$X_t = X_0 + \int_0^t b(X_s, s) ds + \int_0^t \sigma(X_s, s) dW_s$$

● $b(X_t, t)$ — drift

Deterministic part: the mean direction of motion.

● $\sigma(X_t, t)$ — diffusion

Random part: the strength of the noise.

● solution $X_t(\omega)$

Each ω gives one random sample path.

Brownian Motion & White Noise

the source of randomness

Brownian motion W_t

- Continuous everywhere, but nowhere differentiable.
- Gaussian process with independent increments.

$$dW_t := W_{t+dt} - W_t \sim \mathcal{N}(0, dt)$$

$$W_t - W_s \sim \mathcal{N}(0, t - s), \quad \mathbb{E}[(W_t - x)(W_s - x)] = \min(t, s)$$

White noise ξ_t

Formal time-derivative of Brownian motion:

$$\xi_t := \frac{dW_t}{dt}$$

$$\mathbb{E}[\xi_t] = 0, \quad \mathbb{E}[\xi_t \xi_s] = \delta(t - s)$$

$$\int_t^{t+\Delta t} \xi_s ds = W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)$$

SECTION 03

Examples

Brownian motion with drift & the OU process

03

Example 1 — BM with Drift

the simplest SDE

Brownian motion with constant drift

Integrate directly — constant coefficients:

$$dX_t = \mu dt + \sigma dW_t$$

$$X_t = X_0 + \mu t + \sigma W_t$$

so the marginal law is Gaussian:

$$X_t \sim \mathcal{N}(X_0 + \mu t, \sigma^2 t)$$

● μ — drift term

Shifts the mean linearly in time: $\text{mean} = X_0 + \mu t$.

● σ — diffusion strength

Variance grows linearly: $\text{Var} = \sigma^2 t$ (spreads out).

● intuition

Deterministic straight-line motion, blurred by noise that widens over time.

Example 2 — OU Process

mean-reverting diffusion

Ornstein–Uhlenbeck process

Drift pulls X_t back to 0 (compare the deterministic ODE):

$$dX_t = -\theta X_t dt + \sigma dW_t \quad (\theta > 0)$$

$$\dot{X}_t = -\theta X_t \Rightarrow X_t = X_0 e^{-\theta t} \rightarrow 0$$

Trick: let $Y_t = e^{\theta t} X_t$ and apply Itô $\Rightarrow$

$$d(e^{\theta t} X_t) = \sigma e^{\theta t} dW_t$$

$$X_t = X_0 e^{-\theta t} + \sigma \int_0^t e^{\theta(s-t)} dW_s$$

● mean-reverting

The $-\theta X_t$ drift always pushes the state back toward the origin.

● OU = noised diffusion

This is the forward process behind variance-preserving diffusion models: an explicit solution as a decayed start plus an Itô integral of noise.

SECTION 04

Transition Distribution

Marginal & conditional laws

04

Moments of the OU Process

mean, variance, stationary law

Mean & variance (via Itô isometry)

$$\mathbb{E}[X_t] = X_0 e^{-\theta t} \longrightarrow 0$$

$$\text{Var}(X_t) = \sigma^2 \int_0^t e^{-2\theta(t-s)} ds = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t})$$

Combining the two gives the full marginal law:

$$X_t \sim \mathcal{N}\left(X_0 e^{-\theta t}, \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t})\right) \longrightarrow \mathcal{N}\left(0, \frac{\sigma^2}{2\theta}\right)$$

Stationary distribution

As $t \rightarrow \infty$ the mean decays to 0 and the variance saturates at $\sigma^2/2\theta$.

The process forgets its start and converges to a fixed Gaussian — **the equilibrium the diffusion relaxes to.**

Transition Distribution

the law of X_t given X_s

Conditional law (Markov property)

Starting from X_s , only the noise after time s matters:

$$X_t = e^{-\theta(t-s)} X_s + \sigma \int_s^t e^{-\theta(t-u)} dW_u$$

$$\mathbb{E}[X_t \mid X_s = x] = x e^{-\theta(t-s)}$$

$$\text{Var}(X_t \mid X_s = x) = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta(t-s)})$$

Transition density $N(m, V)$:

$$p(s, x; t, y) = \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{(y-m)^2}{2V}\right)$$

● Markov property

The future depends on the present only — not the full past.
The first term is deterministic in X_s ; the rest is fresh Brownian noise.

● SDE → diffusion process

A single SDE (an equation) defines a whole family of transition kernels — the diffusion process X_t . This kernel is what we learn to reverse.

Why “Diffusion”?

local drift & spread

Small-time increment

Over a short step Δt the update splits into a deterministic push plus a random kick:

$$X_{t+\Delta t} - X_t \approx b(t, X_t) \Delta t + \sigma(t, X_t) \Delta W_t$$

Local drift

$$\mathbb{E}[\Delta X \mid X_t = x] \approx b(t, x) \Delta t$$

The conditional mean move — how the center of mass drifts.

Random diffusion

$$\text{Var}(\Delta X \mid X_t = x) \approx \sigma^2(t, x) \Delta t$$

The conditional variance — how the probability mass spreads out (diffuses).

SECTION 05

Generator & Fokker–Planck

How the density evolves

05

Infinitesimal Generator

how $E[f(X_t)]$ evolves

From Itô's formula to the generator

Apply Itô to $f(X_t, t)$:

$$df = \left(\partial_t f + b \partial_x f + \frac{1}{2} \sigma^2 \partial_{xx} f \right) dt + \sigma \partial_x f dW_t$$

Collect the drift part into the generator $\mathcal{L}$:

$$\mathcal{L}f = b \partial_x f + \frac{1}{2} \sigma^2 \partial_{xx} f$$

Taking expectations kills the martingale (dW) term:

$$\frac{d}{dt} \mathbb{E}[f(X_t)] = \mathbb{E}[(\mathcal{L}f)(X_t)]$$

● computing rules

$$(dt)^2 = dt dW_t = 0, \quad (dW_t)^2 = dt$$

● $\mathcal{L}$ = generator

$\mathcal{L}$ is the operator describing how the average of any test function f is transported by the process — the infinitesimal engine of the dynamics.

Fokker–Planck Equation

the density's evolution

Adjoint of the generator

Move $\mathcal{L}$ onto the density p by integration by parts (adjoint $\mathcal{L}^*$):

$$\int (\mathcal{L}f) p \, dx = \int f (\mathcal{L}^*p) \, dx$$

gives the 1-D Fokker–Planck (forward Kolmogorov) equation:

$$\partial_t p = \mathcal{L}^* p = -\partial_x (b p) + \frac{1}{2} \partial_{xx} (\sigma^2 p)$$

For $dX_t = f \, dt + g(t) \, dW_t$ (vector form):

$$\partial_t p = -\nabla \cdot (f p) + \frac{1}{2} g^2 \Delta p$$

● what it says

A PDE for the whole density $p_t(x)$: drift transports mass, diffusion spreads it. The bridge from a single path to the evolving distribution.

OU example

$$\partial_t p = \theta \partial_x (x p) + \frac{\sigma^2}{2} \partial_{xx} p$$

SECTION 06

Probability-Flow ODE

A deterministic sampler & the score

06

Probability-Flow ODE

matching the marginals

Find a deterministic flow with the same marginals

Seek a deterministic ODE $\dot{x} = v(x, t)$ whose density obeys the continuity equation:

$$\partial_t p_t = -\nabla \cdot (p_t v)$$

Rewrite the diffusion term as a divergence (a score flux):

$$\frac{1}{2} g^2 \Delta p = \nabla \cdot \left(\frac{1}{2} g^2 p \nabla \log p \right)$$

Substitute into Fokker–Planck and match the two divergence forms:

$$\partial_t p = -\nabla \cdot \left(f p - \frac{1}{2} g^2 p \nabla \log p \right) = -\nabla \cdot (p v)$$

Reading it

A stochastic (SDE) and a deterministic (ODE) process can share the **exact same time-marginals p_t** — the noise is traded for a drift correction, provided we know $\nabla \log p_t$.

The Score Function

the payoff

Probability-flow velocity field

$$v(x, t) = f(x, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x)$$

i.e. the equivalent deterministic sampler is:

$$\frac{dX_t}{dt} = f(X_t, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(X_t)$$

The score function

$$s(x, t) := \nabla_x \log p_t(x)$$

The gradient of the log-density. The one unknown the ODE needs — everything else is given by the SDE.

Why we care

Learn the score with a neural net $\Rightarrow$ **score-based diffusion models**. The PF-ODE gives fast, deterministic, invertible sampling with the same distribution as the SDE.

Thank you



Jangha Lee

Perception & Physical Intelligence Lab | Seoul National University